

Statistical Methods for Experimental Physics (Part-I)

Problem Set

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1 Frequentist Statistics Problems

These problems are entirely optional and will not be assessed. You might find them useful to prepare for the Exam or to check your understanding of the material.

1.1 Kolmogorov axioms

Show that the definition of *frequentist* probability given in the lectures satisfies the three Kolmogorov axioms (I don't expect any very rigorous proofs). For this question, you can always assume we are concerned with the probabilities related to *exclusive outcomes*.

Extra Challenge If you can also show this for the Bayesian definition, then have a go!

1.2 Zero correlation does not mean independence

In the lectures, we said that two random variables which are independent will have a zero correlation coefficient.

1. Show that two continuous random variables X and Y , with $(X, Y) \sim f(X, Y)$ which are independent will have a 0 correlation coefficient.
2. Let X be a continuous random variable symmetrically distributed around 0 with a density function $f(X)$. Let $Y = X^2$. Show that despite the fact that Y and X are clearly dependent, their correlation coefficient is 0.

1.3 Moments of the Poisson distribution

In the lecture notes, we calculated the expectation and variance of the Poisson distribution with parameter λ are $E[k] = \lambda$ and $V(k) = \lambda$.

What is the third central moment of the Poisson distribution? You should show how you derive your answer.

1.4 Cauchy distribution

In the lectures, we showed how the sum of two Gaussian distributed random variables is itself Gaussian distributed. Suppose now that $X \sim \phi(X; 0, 1)$ and $Y \sim \phi(Y; 0, 1)$ are independent random variables. Show that the distribution of $U = \frac{X}{Y}$ is Cauchy, i.e that $p(U) = \frac{1}{\pi(1 + U^2)}$.

Hint: You should start by deriving the marginal distribution formula for the ratio of two independent random variables. Be careful that the case $Y = 0$ will cause a problem, so split the marginal distribution into two cases, one for $Y > 0$ and one for $Y < 0$. The sum of these marginal distributions will be the total distribution.

1.5 The Lying Friend

A friend of yours is known to only tell the truth with a probability $0 \leq x \leq 1$.

- i Suppose you have a deck of 52 cards and this friend draws one of the cards and declares that it's a King. What is the probability that the card is a King? Express the answer in terms of x .

- ii Plot this probability as a function of x . What would x need to be before you are more than 50% sure that the card is a King?

Hint: A standard deck of cards contains 4 Kings. You can use Bayes' rule to solve this question.

1.6 Convergence to a uniform distribution

Let X_n be a random variable with probability density function $f_n(X)$ given by,

$$f_n(X) = 1 - \cos 2n\pi X$$

for $X \in [0, 1]$ and $f_n(X) = 0$ otherwise. Show that X_n converges in distribution to a random variable X which is uniformly distributed on $[0, 1]$

1.7 Convergence to a Dirac-delta function

Let $X_n \sim \phi(X; 0, 1/n)$ for each n . Show that X_n converges in distribution to a probability density function which is a Dirac delta function at 0 – $\delta(0)$.

Hint: To show this, show that for any $x < 0$, $F_n(x) \rightarrow 0$, and for any $x > 0$, $F_n(x) \rightarrow 1$, as $n \rightarrow \infty$. Providing an answer using a numerical method is sufficient to get the full marks (feel free to write some code to demonstrate this, and attach plots to prove it to me!).

1.8 The Maxwell-Boltzmann distribution

The Maxwell-Boltzmann distribution describes the distribution of the speed, $v \in [0, +\infty)$, of molecules in a gas at temperature T . Its probability density is given by,

$$f(v; T) = \left(\frac{m}{2\pi kT}\right)^{3/2} 4\pi v^2 e^{-\frac{mv^2}{2kT}},$$

where m is the molecular mass and k is a constant known as the Boltzmann constant. The speed v of any molecule will be a random variable $v \sim f(v; T)$.

- i. Show that the first algebraic moment (the expectation of v) of the Maxwell-Boltzmann distribution μ_1 is given by,

$$\mu_1 = E[v] = 2\sqrt{\frac{2kT}{m\pi}}.$$

You may use the following identity,

$$\int_0^{+\infty} x^3 e^{-bx^2} dx = \frac{1}{2b^2}.$$

- ii. Imagine an experimental apparatus can measure speeds of molecules in a gas v and observes a number n of speeds $\{v_1, v_2, \dots, v_n\}$. The observations are independent and identically distributed random variables distributed as $v_i \sim f(v; T)$.

Show that the sample mean, $\bar{v} = \frac{1}{n} \sum_{i=1}^n v_i$, is an *unbiased estimator* for the first algebraic moment of the the Maxwell-Boltzmann distribution.

1.9 Phone call for Peter Higgs

You are sat at home with your statistics problems staring at you from across the room, and rather than solving them, you decide to procrastinate by dialing random telephone numbers. You punch a phone number in only to have Peter Higgs himself pick up at the other end. What is the probability of picking Professor Higgs' number out of all possible phone numbers with an Edinburgh area code? Is this more or less likely than the discovery of the Higgs boson being an accident due to a large fluctuation of the background?

Hint: You can assume the Higgs discovery was at exactly 5σ (or sometimes said "5 standard deviations"). Convert this to a p -value using a one-sided test as explained in lectures. Also assume the Edinburgh phone number can be any 11 digit number starting with 0131.

1.10 Maximum likelihood for exponential decay

Suppose we want to determine the lifetime (τ) of an unstable particle. The probability for this particle to decay at time t is given by,

$$p(t) = N(\tau)e^{-t/\tau} \quad (1)$$

Suppose then that in the lab, we've made measurements of the decay times (say by measuring the times at which we detect one or more decay products from a source), and have the following observed decay times; $t = 7.44, 4.13, 26.85, 1.42, 3.46, 2.68, 4.1, 4.04, 7.9$ and 12.03 in ns.

- i. Normalise the probability density function (i.e what is $N(\tau)$?)
- ii. What does the log-likelihood function look like for this dataset? Use your favourite plotting program to plot it as a function of τ .
- iii. What is the maximum likelihood (or minimum negative log-likelihood) estimator for τ ? **Hint:** You can calculate this analytically or numerically (or even better, do both and compare answers!).